

Learning without labels

THE ONE THING MEDICINE CANNOT BUY

What a label costs

A class label needs a clinician. A segmentation mask needs a clinician for an hour. A study of **a thousand** images is a large study, and ImageNet has **1.2 million**.

What a hospital already has

Hundreds of thousands of **unlabelled** scans sitting in PACS, each one acquired for care and never annotated for research. The pixels are free; only the opinion is expensive.

So invent a task the data can grade itself on. If the supervision comes from the raw scan, the archive becomes a training set without anyone annotating anything.

The field has already voted: papers using self-supervision for medical image classification grew more than **tenfold** between 2019 and 2021.

What it is

SUPERVISED, AND SELF-SUPERVISED

Traditional supervised learning

Train the model directly on the task you care about, using paired inputs and **ground truth labels** for that task: manual class labels, expert masks, outcomes pulled from the record.

Self-supervised learning

Do *not* train on the task. Invent a **pretext task** whose supervision is generated from the raw data itself, and use it to learn a good **feature encoder** for this kind of data. No external labels are used at this stage. Then fine-tune that encoder on the real task with far fewer labels.

This is transfer learning from Module 2, with the source dataset replaced: instead of pretraining on somebody else's labelled photographs, pretrain on *your own unlabelled* scans.

WHAT ARE THE ADVANTAGES OF SELF-SUPERVISED LEARNING?

Scarce labels

Where annotation is manual, expensive, or needs an expert, the label is the bottleneck and nothing else is. Self-supervision puts to work the data that would otherwise sit unused **for want of a label**.

Data at scale

It turns raw volume into a resource. Meaningful patterns are extracted without supervision, so training **scales with the archive** rather than with how much human effort you can buy.

Versatile

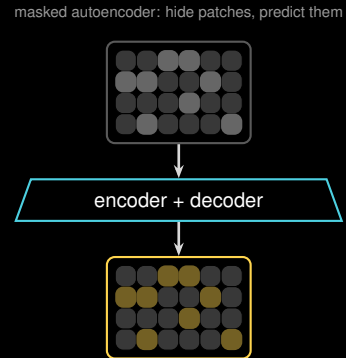
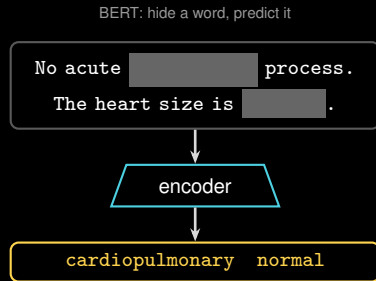
The same recipe learns linguistic structure from text, acoustic structure from speech, and visual structure from images. What comes out is a **general encoder**, not a solution to one task.

Robust, and general

Trained across a broader spectrum of data, the model builds a more complete picture of the underlying patterns and **overfits less** — which is what you need when deployment is diverse and unpredictable.

The first two are the same coin: labels are the scarce thing, unlabelled data is the abundant thing, and self-supervision is the trade between them. The second two are why the field kept going after the labels problem was solved: **it produces better encoders**, not merely cheaper ones. Large language models are the proof at scale.

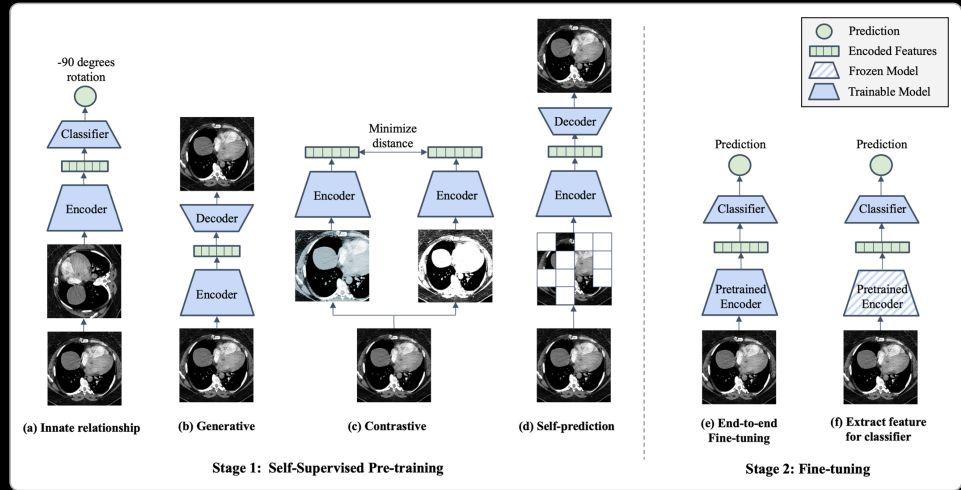
YOU ALREADY USE ONE EVERY DAY



BERT, the network behind web search since 2019, was trained by hiding words and predicting them; GPT predicts the next one. **The sentence is its own label.** The same trick on pixels is a masked autoencoder, and the same trick on a chest x-ray is a pretext task nobody had to annotate.

How it works

FOUR FAMILIES OF PRETEXT TASK



innate relationship

predict the rotation, the jigsaw
15 of 79 studies

generative

reconstruct the whole input
3 of 79

contrastive

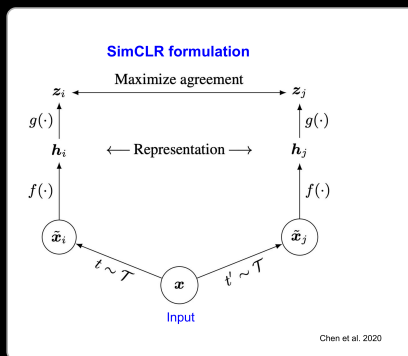
two views should agree
44 of 79

self-prediction

hide a part, reconstruct it
6 of 79

Eleven more studies combined two of them. Contrastive won the field, so this lecture follows it; self-prediction is the challenger, and gets one slide of its own.

SIMCLR: THE SHAPE OF A CONTRASTIVE METHOD



Four moving parts

1. One image x , two random transformations t, t' , giving two views $\tilde{x}_i$ and $\tilde{x}_j$.
2. A **shared** encoder f produces representations h_i, h_j .
3. A small projection head g maps those to z_i, z_j .
4. A loss that pulls z_i and z_j together and pushes every other view in the batch away.

The encoder never learns what the image *is*. It only learns what survives a transformation.

THE CONTRASTIVE LOSS IS CROSS-ENTROPY IN DISGUISE

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau)} \quad \text{sim}(u, v) = \frac{u^\top v}{\|u\| \|v\|}$$

Read it as a classifier

Given view i , its partner j must score higher than all $2N - 2$ other views in the batch. That is a softmax over $2N - 1$ candidates, and the “correct class” is the other view of the same image.

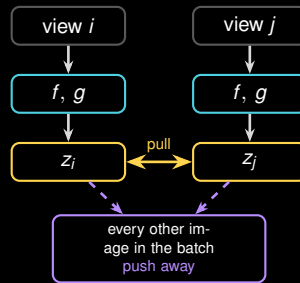
The two knobs

Cosine similarity compares direction, not magnitude. The **temperature** τ sets how peaked the distribution is, and it matters more than it looks.

Same negative log-likelihood we minimized in Module 1. Only the notion of “class” has changed.

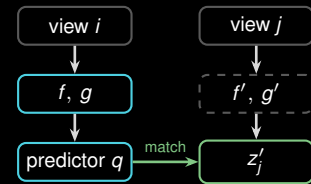
DO YOU EVEN NEED THE NEGATIVES?

SimCLR: contrastive



needs a big batch, or a memory bank

BYOL / SimSiam: positives only

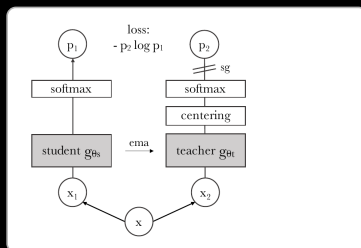


no negatives at all; the right tower is a slow copy of the left, with **no gradient** through it

small batches, no memory bank

Both reach the same accuracy. The invariant is **“two views of one scan should agree”**; the negatives were only ever there to stop the encoder from answering every input with the same vector, and a stop-gradient does that job too. The method is a detail. The invariant is the idea.

DINO: DISTILLATION WITH NO TEACHER TO DISTIL FROM



Two views of x ; the student is trained, the teacher is an exponential moving average of it.

BYOL, with one change

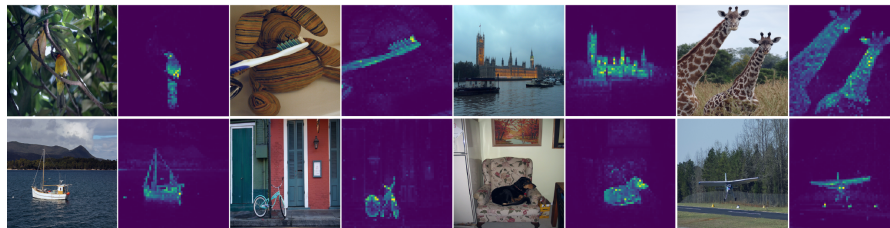
The output is no longer a vector but a **softmax over K prototypes**, and the loss is **cross-entropy** between the teacher's distribution and the student's: $-p_t \log p_s$. Same shape as knowledge distillation, except the teacher is built from the student.

Two knobs against collapse

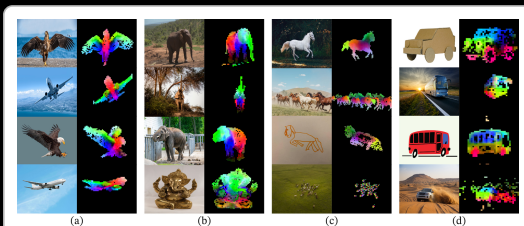
Centering: subtract the batch mean from the teacher's logits, so no single prototype can win everything. **Sharpening**: a low teacher temperature, so it cannot answer "all prototypes equally" either. One pushes towards uniform, the other away; together they hold.

Third time the same loss. SimCLR, the classifier from Module 1, and now DINO all minimize a cross-entropy; only the "correct class" keeps changing. A ViT trained this way with **no labels** reaches **78.3%** on ImageNet with a nearest-neighbour classifier and **80.1%** with a linear one.

WHAT EMERGES WHEN NOBODY SAYS WHAT TO LOOK AT



Self-attention of the class token, last layer, DINO ViT-S/8. No mask, no box, no label was ever shown to this network.

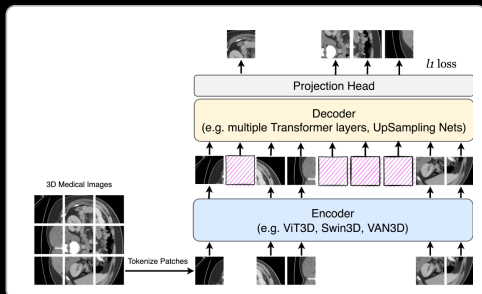


DINOv2: first three PCA components of patch features, one colour each. The same part gets the same colour across images.

Why this matters here

The encoder learned **where the object is** and **which part is which** from the pretext task alone. DINOv2 scaled the recipe to a 1.1B ViT on 142M curated images, then distilled it down; its frozen features are the off-the-shelf encoder much of today's work starts from, medical imaging included.

OR JUST HIDE PATCHES: MASKED IMAGE MODELING



MAE on a 3D CT volume: only the visible patches enter the encoder; a

light decoder fills in the hatched ones.

The recipe

Cut the scan into patches. Hide **75%** of them at random. Encode only the rest, reconstruct the hidden voxels, and take the loss **on the hidden patches only**. Throw the decoder away.

Why hide so much

Images are redundant. Hide 20% and the network copies the neighbours; hide 75% and it has to know where the kidney is to draw it.

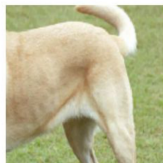
Against contrastive, on CT organ segmentation: MAE and SimMIM reached a Dice of 0.75–0.76 where SimCLR reached 0.72, and hit 0.70 in $1.4\times$ less training time. And note the reversal of the previous slides' worry: here the hidden lesion is the **answer**, not a nuisance to be invariant to.

The medical catch

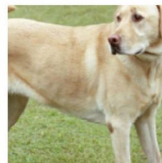
ASK THE ROOM: WHICH OF THESE ARE SAFE ON A CHEST X-RAY?



(a) Original



(b) Crop and resize



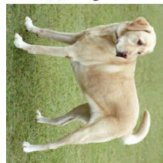
(c) Crop, resize (and flip)



(d) Color distort. (drop)



(e) Color distort. (jitter)



(f) Rotate $\{90^\circ, 180^\circ, 270^\circ\}$



(g) Cutout



(h) Gaussian noise



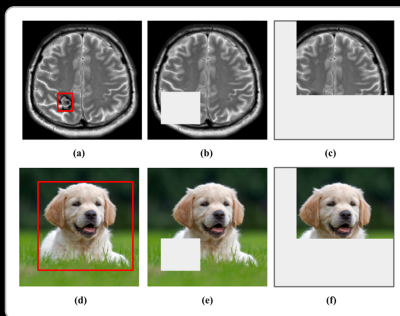
(i) Gaussian blur



(j) Sobel filtering

SimCLR's whole result rests on the transformation set: the encoder learns exactly what those transformations tell it to ignore. Crop and resize, colour distortion, rotation, cutout, Gaussian noise, blur, Sobel filtering. **Which of them would you allow on a radiograph?**

WHICH OF THESE ARE SAFE ON A CHEST X-RAY?



T2 MRI with a cavernoma (red), then masked, then cropped.

The dog survives both. The lesion survives neither.

Usually fine

Small crops, mild blur, mild noise, small rotations: real acquisition variability.

Meaningless

Colour jitter and colour drop on **one channel**. The transformation the paper found most important does not exist here.

Dangerous

Horizontal flip teaches that **left and right do not matter**. Large crops and random masks can **delete the finding**, as on the left.

Two things break SimCLR's assumptions on medical images: scans of one protocol all look **alike**, so the class is not "the main object", and the finding is **small and local**, so a crop that would never lose a dog can lose a tumour. In supervised training a bad augmentation costs accuracy; in a pretext task it defines what the representation is allowed to encode.

BUILD POSITIVES FROM THE CLINIC

adjacent slices
of one CT series

two views
of one study: PA and
lateral, CC and MLO

one patient
scanned again a year later

image + its report
[next lecture](#)

each is a "same thing, different view" pair that already exists in PACS, with no transformation invented by you

Why it is safer

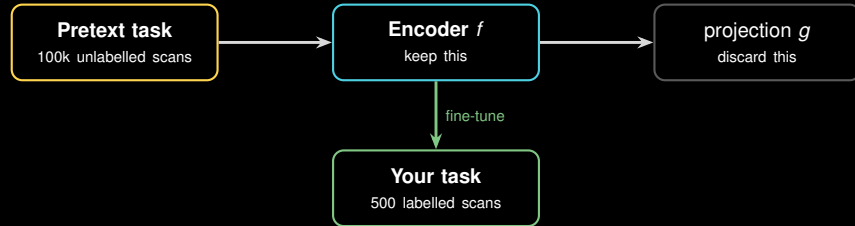
A neighbouring slice or a later scan still contains the lesion. The invariance you teach is "same anatomy, same patient", which is the one you actually want.

Why it works better

Across the review, pretext tasks built on clinical structure improved downstream results by **13.7%** over supervised baselines; generic transformation tasks by **5.0%**.

What it buys

WHAT YOU KEEP, AND HOW TO FINE-TUNE IT



What you carry over

The projection head exists only to give the loss a place to live. What reaches the real task is the encoder, and the payoff is **label efficiency**: matching a fully supervised model with a fraction of the annotations.

How you fine-tune

End-to-end: unfreeze the encoder and train everything on your labels. Freezing it and training only a linear head is cheaper, but end-to-end won in **8 of the 11** studies that compared them.

DOES IT WORK? SEVENTY-NINE STUDIES SAY MOSTLY

The gain

+6.35%

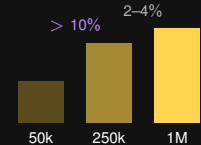
average gain, 36 contrastive studies

Gains ran from almost nothing to **+32.6%** AUROC. Five studies came out **worse**, by 1 to 4.5 points. It is not free.

Which weights

3. supervised ImageNet weights
2. **self-supervised** ImageNet weights: already better on medical tasks
1. those, then **keep pre-training on your own scans**

How much data



Going from 1M to 250k pre-training images costs 2–4 points; 250k to 50k costs more than 10. There is a floor.

Radiology supplied 47 of the 79 studies, 27 of them chest imaging, and SimCLR was the method in 13. When it fails, it fails for the reason on the previous slides: the pretext task taught the encoder to ignore the wrong thing.

A RECIPE

Self-supervision on your own archive, in five steps

1. Start from **public self-supervised weights**, not supervised ImageNet ones. Only 13 of 79 studies did.
2. Keep pre-training on **your own unlabelled scans**. Tens of thousands at the very least; the curve is steep below that.
3. Choose transformations and positive pairs that **cannot delete the finding**: neighbouring slices, paired views, the same patient over time.
4. **Fine-tune end-to-end** on the labels you have.
5. Compare against **at least one other family** of pretext task. Only 33 of 79 studies did, and the winner changes with the modality.

This is the transfer-learning slide from Module 2 with one word changed: the source domain is no longer somebody else's photographs but your own hospital's unlabelled archive.